Evaluating the Transport Impact of Large Post-Quantum DNSSEC Signatures

Theoretical Part (Project – Part 1)

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***Abstract*—Post-quantum signature schemes such as ML-DSA-44, standardized by NIST in FIPS 204, produce signatures and public keys that are one to two orders of magnitude larger than those of the classical algorithms used in DNSSEC today. As a result, signed DNS responses routinely exceed the 1,232-byte EDNS(0) payload recommendation and the typical Internet MTU, which pushes them onto IP fragmentation, truncation with TCP fallback, or alternative transports. This theoretical part reviews the DNS and DNSSEC foundations, the classical and post-quantum signature algorithms relevant to DNSSEC, and the message-size and transport mechanisms that govern how large responses are delivered, including TCP fallback, Application-layer Request-based Resource Record Fragmentation (ARRF), and DNS over QUIC (DoQ). It then surveys related work and identifies the gap addressed by the practical part of the project: a controlled, reproducible comparison of these transports for ML-DSA-44 against an ECDSA baseline under varying latency and packet loss.**

***Index Terms*—DNSSEC, post-quantum cryptography, ML-DSA, TCP fallback, IP fragmentation, ARRF, DNS over QUIC**

I. Introduction

The Domain Name System (DNS) translates human-readable domain names into the numerical addresses that are required for almost every transaction on the Internet [1]. Despite this critical role, the original DNS protocol was not designed with any mechanism for verifying that a response actually originates from the authoritative source or that it has not been altered in transit [2]. The DNS Security Extensions (DNSSEC) were introduced to close this gap by adding data origin authentication and data integrity protection to DNS responses through digital signatures, without, however, providing confidentiality [2]–[4]. More than two decades after its initial specification, DNSSEC is considered the best current practice for authenticating DNS data [5], even though deployment remains uneven: recent estimates suggest that fewer than ten percent of website domain names are signed and that only around a third of queries to recursive resolvers are actually validated [5].

The cryptographic foundation of DNSSEC, however, is now under long-term threat. The signature algorithms that are most widely deployed today, in particular RSA and the Elliptic Curve Digital Signature Algorithm (ECDSA) [6], [7], derive their security from the computational hardness of integer factorization and the discrete logarithm problem. A sufficiently large, cryptographically relevant quantum computer would be able to solve both of these problems efficiently using Shor's algorithm [8], which would allow an attacker to forge DNSSEC signatures and therefore to present spoofed DNS data as authentic.[1] To address this risk, the U.S. National Institute of Standards and Technology (NIST) has standardized the Module-Lattice-Based Digital Signature Algorithm (ML-DSA) in FIPS 204, a signature scheme whose security relies on the hardness of lattice problems and that is believed to remain secure even against an adversary equipped with a quantum computer [9].

Post-quantum cryptography (PQC) schemes such as ML-DSA, however, do not come without cost. The smallest standardized ML-DSA parameter set, ML-DSA-44, produces signatures of 2,420 octets and public keys of 1,312 octets [9], [10], which is one to two orders of magnitude larger than a typical ECDSA signature. For comparison, an ECDSA P-256 signature occupies only 64 bytes, so an ML-DSA-44 signature is almost 38 times larger [11]. When such a signature is embedded into a DNSSEC response together with the DNS

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[1]This is a long-term, not an immediate, threat: at the time of writing, no cryptographically relevant quantum computer is publicly known to exist. Nevertheless, migration planning for long-lived infrastructure such as DNS is widely recommended well in advance of such a scenario.

header, resource records, and—where applicable—a DNSKEY record, the resulting message can easily exceed the conservative 1,232-byte UDP payload size that is commonly recommended in operational practice today, and it will frequently exceed the roughly 1,500-byte Maximum Transmission Unit (MTU) of a typical Ethernet path [12], [13]. Because DNS was originally designed around a 512-byte UDP message size [1] and later extended via EDNS(0) to negotiate larger UDP payloads [12], oversized post-quantum DNSSEC responses risk falling back to one of two imperfect mechanisms: IP-layer fragmentation, which is unreliable because many firewalls and other middleboxes silently drop fragmented packets [14], [15], or truncation of the UDP response followed by a retry over TCP, which guarantees delivery but adds at least one additional round trip and extra server-side state [14]. A drop-in replacement of classical signatures with quantum-safe ones is expected to increase DNSSEC query resolution times by at least a factor of two: a response carrying large post-quantum signatures is likely to be truncated, which wastes the initial UDP round trip, and the subsequent retry over TCP incurs at least two further round trips because of the three-way handshake [16].

The DNS operator community has already begun to experiment with mitigations for this problem. Cloudflare has deployed experimental ML-DSA-44 validation on its public 1.1.1.1 resolver, relying on TCP fallback whenever a response becomes oversized [11], and an open-source extension of the CoreDNS authoritative server has demonstrated on-the-fly signing with several PQC signature families, including ML-DSA [17], [18]. At the same time, the research literature has proposed alternative transport mechanisms that try to avoid the drawbacks of both IP fragmentation and plain TCP fallback. Application-layer Request-based Resource Record Fragmentation (ARRF) fragments oversized DNS resource records at the DNS protocol layer itself rather than relying on the IP layer, with the resolver requesting any missing fragments explicitly [19]. DNS over QUIC (DoQ) instead moves the entire DNS exchange onto a dedicated, encrypted QUIC connection, which has its own packetization and loss-recovery behavior and therefore sidesteps classical IP fragmentation entirely [20].

Although each of these mechanisms—plain TCP fallback, ARRF, and DoQ—has been described or prototyped individually, a direct, controlled, and reproducible comparison of their latency, bandwidth, and reliability behavior specifically under the conditions created by post-quantum DNSSEC signatures is still largely missing from the public literature. This gap motivates the present project. Using ECDSA-based DNSSEC as a classical baseline and an ML-DSA-44-based DNSSEC deployment as the post-quantum case, this project evaluates DNS resolution under simulated network conditions with varying latency and packet loss, measuring query latency, resource utilization, bandwidth consumption, packet sizes, and the frequency of TCP fallback. Time permitting, the evaluation is extended to ARRF-based fragmentation and to DNS over QUIC, in order to assess which combination of signature algorithm and transport mechanism currently represents the most realistic path toward a quantum-resistant DNSSEC deployment.

Concretely, the project is guided by the following research questions, which will be answered in the practical part (Project – Part 2):

- **RQ1:** How much do ML-DSA-44 signatures increase DNSSEC response sizes, packet sizes, and bandwidth consumption, and how often do they trigger truncation and TCP fallback, compared with the ECDSA baseline?
- **RQ2:** How do the end-to-end query latency and resource utilization of post-quantum DNSSEC with TCP fallback change as path latency and packet loss increase?
- **RQ3:** (Time permitting.) To what extent can ARRF or DNS over QUIC reduce the overhead identified in RQ1 and RQ2?

The remainder of this (theoretical) part of the report is structured as follows. Section II introduces the theoretical foundations that are necessary to understand the problem: the DNS and DNSSEC protocols, the classical and post-quantum signature algorithms relevant to this project, and the mechanisms that govern DNS message size and transport selection. Section III positions this project with respect to the existing state of the art and related work. Section IV closes the theoretical part with a brief summary and a transition toward the practical implementation and evaluation that are reported in Part 2 of this project. The sources of this report and its figures, and, in Part 2, the testbed configuration, measurement scripts, and results, are publicly available in the project repository at https://github.com/sezRR/pqc-dnssec-research.

II. Theoretical Foundations

A. The Domain Name System and Its Security Model

The Domain Name System provides a globally distributed, hierarchical naming service that maps domain names to resource records, most commonly IP addresses [1]. A DNS resolver issues a query for a given name and record type and receives a response from an authoritative name server, typically via a recursive resolver that caches intermediate results. In its original specification, DNS offers no cryptographic guarantee that a received response was actually produced by the zone's authoritative server or that it was not modified or injected by an on-path or off-path attacker [2]. This absence of origin authentication is the central weakness that DNSSEC was designed to address.

B. DNS Security Extensions (DNSSEC)

1) Core Resource Records and the Chain of Trust: DNSSEC, as specified across RFC 4033, RFC 4034, and RFC 4035, extends DNS with a small set of new resource record types and protocol behaviors that together provide data origin authentication and data integrity [2]–[4]. A zone owner signs its resource record sets (RRsets) with a private key, producing RRSIG records that contain the resulting digital

signature; the corresponding public key is published in a DNSKEY record. A DS (Delegation Signer) record, stored in the parent zone, contains a cryptographic hash of the child zone's DNSKEY and thereby links the child's key material into the chain of trust that ultimately terminates at the signed DNS root. A validating resolver can therefore follow this chain from the root down to the queried name, verifying each signature along the way, to establish that a response is authentic. Importantly, DNSSEC does not encrypt DNS traffic and therefore provides no confidentiality; its guarantees are limited to authenticity and integrity [2].

RFC 4033 additionally distinguishes two roles for zone keys: a Key Signing Key (KSK) corresponds to a private key used to sign one or more other authentication keys of a zone, whereas a Zone Signing Key (ZSK) corresponds to a private key used to sign the zone itself; typically, the KSK signs the DNSKEY RRset containing the ZSK, and the ZSK signs the other zone data [2]. This distinction is purely operational: validation does not differentiate between the two roles, and a single key may serve as both [2]. The DS RRset in the parent zone indicates the key(s) used to self-sign the child's DNSKEY RRset, so the typical authentication chain is DNSKEY $\rightarrow$ [DS $\rightarrow$ DNSKEY]* $\rightarrow$ RRset [2]. Fig. 1 illustrates how DS, DNSKEY, and RRSIG records combine into the chain of trust [2]–[4]. Two properties of this design are directly relevant for this project. First, when a query has the DO bit set, a security-aware authoritative server should send the RRSIG records needed to authenticate the RRsets in the response and keep each RRset together with its RRSIG(s) [4], while the DNSKEY RRset carries the complete public keys of the zone [3]; the sizes of signatures and keys therefore translate directly into the size of DNS messages. Second, a sender must not use DNS name compression on the Signer's Name field of an RRSIG record, so this field always contributes its full length [3].

2) Adoption and Current Deployment Status: Although DNSSEC is regarded as the current best practice for DNS origin authentication, and although the DNS root zone and many top-level domains (TLDs) are signed, end-to-end deployment below the TLD level remains limited: current estimates indicate that fewer than ten percent of website domain names are DNSSEC-signed and that only roughly a third of queries handled by recursive resolvers are actually validated [5]. This uneven deployment is relevant to the present project because any additional transport overhead introduced by post-quantum signatures could further discourage adoption if it is not kept within acceptable latency and reliability bounds. Support for TCP, the fallback transport, is also not universal: according to measurements published in 2016 and cited by the authors of [21], about 11% of name servers lacked TCP support [21].

C. Cryptographic Signature Algorithms and the Quantum Threat

1) Classical Signature Algorithms Used in DNSSEC: The DNSSEC specifications allow multiple signature algorithms to be registered and used, and operational guidance on which algorithms should be implemented and used for signing or validation is maintained separately from the core protocol documents [7]. In current practice, RSA and ECDSA are the most widely deployed choices [6], [7]. ECDSA in particular is attractive for DNS because it produces comparatively small signatures and public keys, which keeps DNSSEC-signed responses close to the message sizes that classical, unsigned DNS traffic already uses. For this reason, ECDSA is used as the classical baseline algorithm in this project.

Concretely, an ECDSA P-256 (DNSSEC algorithm 13) signature is encoded as the concatenation $r \,|\, s$ of two 32-octet integers, i.e., 64 octets [6], and the public key is the uncompressed curve point $x \,|\, y$, which occupies 64 bytes [6], [11]; RSA-2048 with SHA-256 (algorithm 8) uses 256-byte signatures and 260-byte public keys [11]. The Edwards-curve algorithm Ed25519 is even more compact, with 32-octet public keys and 64-octet signatures [22]. All of these values remain far below the 1,232-byte UDP payload limit discussed in Section II-D1.

2) The Quantum Threat to Classical Public-Key Cryptography: Both RSA and ECDSA rely on mathematical problems—integer factorization and the elliptic-curve discrete logarithm problem, respectively—that are believed to be intractable for classical computers at the key sizes used in practice. A sufficiently large quantum computer running Shor's algorithm [8] would, however, be able to solve both problems in polynomial time. If such a quantum computer became available, an attacker could use it to derive the private signing key of a DNSSEC zone (or otherwise forge valid-looking signatures) from publicly available information, defeating the authentication guarantees that DNSSEC is meant to provide. Because DNS is expected to remain a core piece of Internet infrastructure for the foreseeable future, and because there is currently no widely deployed alternative to it, this motivates investigating post-quantum alternatives for DNSSEC signing well ahead of the point at which such an attack becomes practical [9].

3) Post-Quantum Cryptography and the ML-DSA Standard: In August 2024, NIST finalized FIPS 204, the Module-Lattice-Based Digital Signature Standard (ML-DSA), as one of the first standardized post-quantum digital signature algorithms [9]. ML-DSA's security is based on the conjectured hardness of problems over module lattices, specifically variants of the Module Learning With Errors and Module Short Integer Solution problems, for which no efficient quantum algorithm is currently known. FIPS 204 defines three parameter sets, ML-DSA-44, ML-DSA-65, and ML-DSA-87, offering increasing security levels at increasing signature and key sizes [9]. An Internet-Draft from the DNS Operations working group specifies how ML-DSA-44 can concretely be carried in DNSSEC's DNSKEY and RRSIG resource records, and states that an ML-DSA-44 public key occupies 1,312 octets while a corresponding signature occupies 2,420 octets [10]. ML-DSA-44 is chosen in that document, and consequently also in this project, because it offers the smallest keys and signatures among the three ML-DSA parameter sets while still targeting NIST

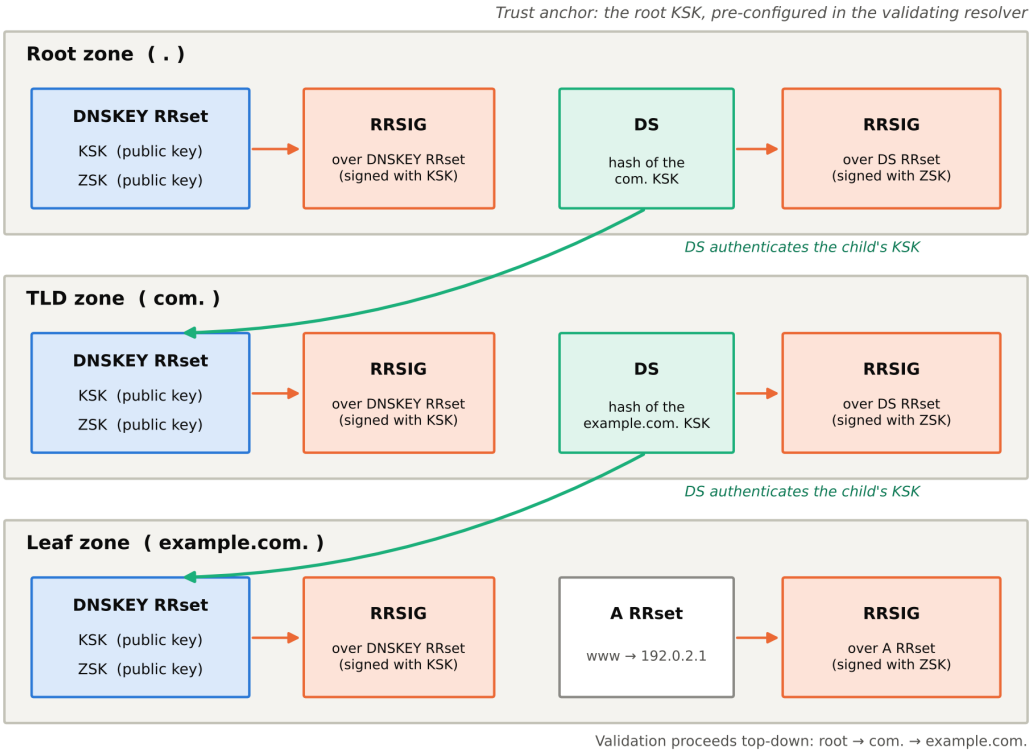


Fig. 1. Simplified DNSSEC chain of trust with the typical KSK/ZSK split.

security category 2, making it the most DNS-friendly of the standardized options [10]. This Internet-Draft is, at the time of writing, still a work in progress and has not yet been published as an RFC; it is used here as the most concrete and current specification of how ML-DSA integrates with DNSSEC, but its details may still change before final standardization.

Table I and Fig. 2 place ML-DSA-44 in the context of other classical and post-quantum signature algorithms; the sizes are taken from [6], [9]–[11], [17], [22], and the thresholds marked in Fig. 2 are the 1,232-byte EDNS(0) recommendation [23] and the ≈1,500-byte Internet MTU [13]. FIPS 204 states that ML-DSA is derived from CRYSTALS-Dilithium, one of the schemes selected in the NIST post-quantum standardization process [9]. Among the post-quantum schemes integrated into CoreDNS in [17], Falcon-512 has a considerably smaller signature (752 bytes) than ML-DSA-44 (2,420 bytes), while SPHINCS+-128s combines a very small 32-byte public key with a 7,856-byte signature [17]. The same study reports that all evaluated post-quantum schemes exceed typical UDP size limits and therefore force the use of TCP, whereas the classical algorithms stay within these limits [17].

The authors of [21] formulated explicit requirements for DNSSEC signature algorithms, the most important being that a signature should not exceed 1,232 bytes; further requirements are a validation speed of at least 1,000 signatures per second and a signing speed of at least 100 signatures per second; public keys of up to 64 kB are considered good, whereas larger keys are accepted only conditionally [21]. Every ML-DSA parameter set already violates the first requirement on signature size alone, which motivates the focus of this project on how the resulting large responses are transported rather than on the choice of signature algorithm alone.

TABLE I
PUBLIC-KEY AND SIGNATURE SIZES OF SIGNATURE ALGORITHMS RELEVANT TO DNSSEC

Algorithm	Family	Public key (B)	Signature (B)
Ed25519 [22]	Classical (elliptic curve)	32	64
ECDSA P-256 [6], [11]	Classical (elliptic curve)	64	64
RSA-2048/SHA-256 [11]	Classical (RSA)	260	256
Falcon-512 [17]	Post-quantum (lattice)	897	752 (666 padded)
ML-DSA-44 [9], [10], [17]	Post-quantum (lattice)	1,312	2,420
ML-DSA-65 [17]	Post-quantum (lattice)	1,952	3,309
ML-DSA-87 [17]	Post-quantum (lattice)	2,592	4,627
SPHINCS+-SHA2-128s [17]	Post-quantum (hash-based)	32	7,856

D. DNS Message Size Constraints and Transport Behavior

1) UDP Payload Limits and EDNS(0): DNS was originally specified to limit messages transported over UDP to 512 octets [1]. The Extension Mechanisms for DNS, known

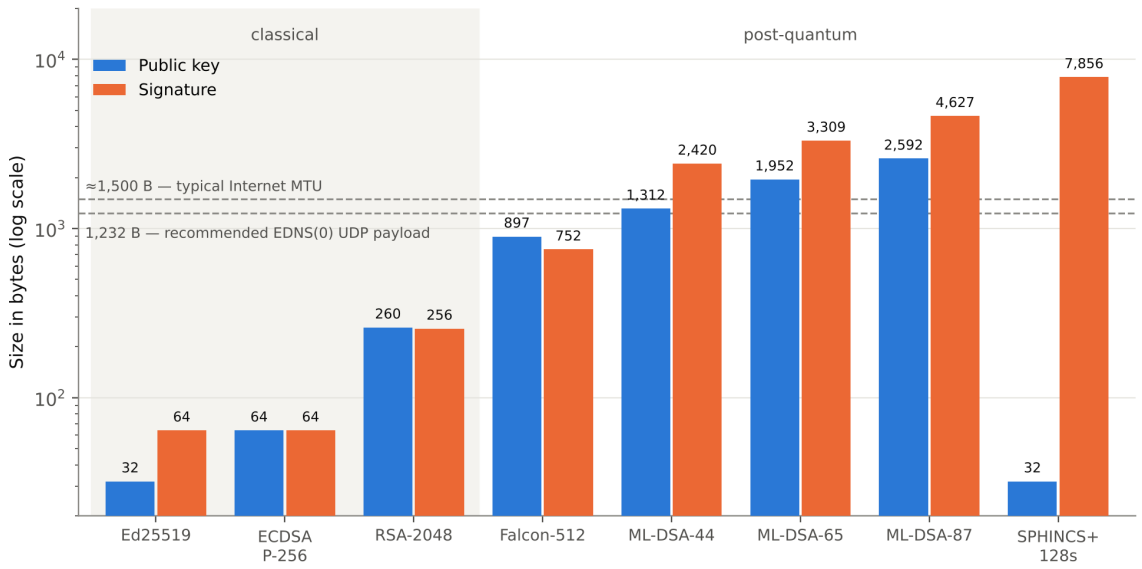


Fig. 2. Public-key and signature sizes (logarithmic scale) compared with the 1,232-byte EDNS(0) recommendation and the ≈1,500-byte Internet MTU.

as EDNS(0), were introduced to let a requestor advertise a larger UDP payload size that it is able to receive, allowing responders to send bigger UDP responses—for example, to carry DNSSEC signatures or multiple address records—without immediately falling back to TCP [12]. EDNS(0) does not, however, remove the underlying physical limits of the network path: RFC 6891 explicitly recommends choosing an EDNS buffer size with care, because advertising a very large value will reliably trigger IP-layer fragmentation, whereas advertising too small a value increases how often TCP is required [12]. In current operational practice, resolvers and authoritative servers frequently converge on a conservative EDNS buffer size of around 1,232 bytes, chosen so that a UDP response together with typical IPv6 and UDP headers still fits inside a common path MTU without fragmentation [12], [13].

The widely used 1,232-byte value is based on the minimum MTU of 1,280 bytes [23], which the IPv6 specification requires for every link in the Internet [24]; subtracting the 40-byte IPv6 header and the 8-byte UDP header gives the largest DNS payload that fits into such a packet [16]:

$$1232 = 1280 - 40 - 8. \tag{1}$$

More recently, RFC 9715 recommends a maximum DNS/UDP payload size of 1,400 bytes as a default, while explicitly allowing smaller values such as 1,232 bytes [23]. In either case, a response of size S can only be sent in a single UDP datagram if it does not exceed the applicable payload limit B, which is determined by the UDP payload size advertised by the requestor [12] and the responder's own limits [23]; a response that does not fit is truncated and the TC flag is set (Section II-D3) [1], [12].

2) IP Fragmentation and Its Reliability Problems: When a UDP datagram exceeds the path MTU—commonly around 1,500 bytes for an Ethernet-based path [13]—it must be split into multiple IP fragments for transmission and reassembled by the receiver. This fragmentation is known to be unreliable in the public Internet: many firewalls and other middleboxes block fragmented IP packets outright, some devices do not implement fragment reassembly correctly, and fragmentation has additionally been shown to open cache-poisoning attack surfaces, because an attacker who can inject a single matching fragment may be able to manipulate the reassembled DNS message [14], [15]. Because large post-quantum DNSSEC responses are substantially more likely to exceed the path MTU than classical, ECDSA-signed responses, they are correspondingly more exposed to these fragmentation-related failure modes [14], [25]. RFC 9715 goes further and states that a DNS message receiver cannot trust fragmented UDP datagrams, recommending that requestors drop fragmented DNS/UDP responses without IP reassembly to avoid cache-poisoning attacks [23]. The authors of [21] additionally note, referring to a 2014 study [25], that up to 10% of resolvers may be unable to handle fragmented responses at all [21], [25].

3) Truncation and Fallback to TCP: When a DNS server cannot fit a response into the UDP payload size negotiated via EDNS(0), it truncates the response and sets the TC (truncated) flag in the DNS header; a compliant client is expected to interpret this flag as an instruction to retry the same query over TCP [12], [14]. TCP avoids the reliability problems of IP fragmentation, because TCP stacks generally perform Path MTU Discovery and segment data accordingly, and because a successful TCP handshake already implies that packets of that size can traverse the path [14]. This reliability, however, comes at a cost: establishing a new TCP connection adds at least one additional round-trip time (RTT) compared to a single UDP exchange, and it requires both client and server to maintain additional connection state [14]. RFC 7766 notes that this

overhead can be partially amortized through connection reuse and query pipelining, but for DNS resolution scenarios where only a single query is being answered—which is common for DNSSEC lookups—the additional round trip is largely unavoidable [14].

The authors of [16] describe the resulting message flow in detail: the initial UDP exchange is wasted because it only returns a truncated response, after which the TCP three-way handshake and the query and response over TCP add at least two further round trips [16]. For a single query on a new connection, an oversized response therefore costs one wasted UDP round trip followed by at least two round trips over TCP [16].

This is consistent with the round-trip counts reported in [26], whose authors expect one RTT for DNS over UDP and two RTTs (handshake plus query) for DNS over TCP when TCP is used directly [26]. A further round trip can be added when a response does not fit into the TCP initial congestion window: the authors of [16] use an initial window of 10 segments with a maximum segment size (MSS) of 1,220 bytes [16]:

$$initcwnd = 10 \times 1220\,\mathrm{B} = 12.2\,\mathrm{kB}. \tag{2}$$

They observe that SPHINCS+-128s and Dilithium-5 responses exceed this window and therefore need one round trip more than Falcon-512 or Dilithium-2 responses [16]. Packet loss affects large responses in a similar way: for a packet-loss rate of 1% and SPHINCS+-128s, where a resolution session comprises about 46 DNS packets, the authors of [16] estimate the probability that at least one packet is lost as [16]

$$\Pr = 1 - (0.99)^{46} \approx 0.37 \tag{3}$$

and conclude that such a session needs an additional round trip with a probability of about one third [16]. Since larger responses consist of more packets, larger post-quantum responses are expected to be more sensitive to packet loss—one of the effects evaluated in the practical part of this project.

4) Size of Signed DNS Responses: The wire formats of DNS and DNSSEC records explain why key and signature sizes translate directly into message sizes. Every resource record consists of an owner name, the fixed-size fields TYPE (2 bytes), CLASS (2 bytes), TTL (4 bytes), and RDLENGTH (2 bytes), and its RDATA [1]. The RDATA of an RRSIG record consists of fixed-size fields—type covered (2 bytes), algorithm (1 byte), labels (1 byte), original TTL (4 bytes), signature expiration (4 bytes), signature inception (4 bytes), and key tag (2 bytes)—followed by the uncompressed signer's name and the complete signature; the RDATA of a DNSKEY record consists of flags (2 bytes), protocol (1 byte), and algorithm (1 byte), followed by the complete public key [3]. Every RRSIG therefore carries a full signature and every DNSKEY record a full public key.

Cloudflare's published example illustrates the result: an ML-DSA-44-signed response from a test zone was 2,563 bytes long—above both the 1,232-byte recommendation [23] and the 1,500-byte Internet MTU [13]—and was retrieved over TCP after truncation [11]. Negative answers are larger still, as responses for non-existent domains can contain three or more signatures [21].

5) Amplification and Denial-of-Service Considerations: Response size is a security concern as well as a performance one. In reflection-based distributed denial-of-service (DDoS) attacks, the attacker exploits the fact that small DNS queries can trigger much larger responses; the authors of [27] quantify this with the amplification factor, defined as the ratio of response size to query size [27]:

$$A = \frac{S_{\text{response}}}{S_{\text{query}}}. \tag{4}$$

In their large-scale measurement, DNSSEC-signed domains produced an average amplification factor of about 47 for ANY queries, compared with about 6 for unsigned domains, although for other common query types the picture was more nuanced [27]. These measurements date from 2014; since then, RFC 8482 has allowed DNS servers to answer queries with QTYPE=ANY with minimal-sized responses, noting that ANY queries are frequently used to exploit the amplification potential of DNS servers and that small responses make servers less attractive amplifiers [28]. The authors of [21] likewise list the attractiveness of large DNSSEC records for DDoS amplification as a reason why signatures should remain small [21]. Since post-quantum signatures substantially increase the response size (Section II-D4), the transport mechanism also determines how much of this potential is exposed: the TCP handshake generally prevents address spoofing and thus the reflection/amplification attacks that affect UDP [14], and DoQ implementations must limit the worst-case amplification to a factor of 3 [20].

E. Transport Mechanisms for Large Post-Quantum DNSSEC Responses

1) TCP Fallback for Oversized Responses: The most directly deployable mitigation for oversized post-quantum DNSSEC responses is to rely on the existing truncation-and-TCP-fallback mechanism described above: a resolver first attempts the query over UDP and, if the response is truncated, repeats the query over TCP [12], [14]. This is also the approach used in Cloudflare's experimental deployment of ML-DSA-44 validation on its 1.1.1.1 resolver [11] and in an independent open-source implementation that extends the CoreDNS authoritative server with on-the-fly PQC signing [17], [18]. Its main advantage is that it requires no changes to the core DNS wire protocol; its main disadvantage is the additional round trip and connection state associated with every oversized query, which this project measures explicitly.

2) Application-Layer Request-Based Fragmentation (ARRF): As an alternative to both IP fragmentation and TCP fallback, the authors of [19] propose ARRF, which fragments oversized resource records at the DNS protocol layer itself rather than at the IP layer [19]. In this scheme, an initial UDP response contains only a truncated fragment of the oversized record together with an indication that further fragments exist;

the requestor then issues explicit follow-up queries to retrieve the remaining fragments, which are reassembled once all parts have arrived. Because fragmentation and reassembly happen above the transport layer, ARRF avoids the specific reliability problems associated with IP-layer fragmentation, while still allowing the overall exchange to remain UDP-based for each individual request [19]. Its trade-off is a potentially higher number of round trips compared to a single TCP connection, depending on how many fragments are required. In more detail, the responder replaces oversized records in place by RRFRAG pseudo-records, which carry an identifier of the fragmented record, the byte offset and size of the fragment, and the total record size; this first response acts as a "map" of the complete message and sets the TC flag for backward compatibility [19]. The requestor then asks for the missing fragments either sequentially or, in the parallel ("batched") mode, all at once after the map has arrived [19]. For an original response of size S and a maximum message size M, the authors of [19] give the number of round trips required by ARRF as

$$R_{\text{ARRF}} = \left\lceil \frac{S}{M} \right\rceil . \tag{5}$$

With the exception of the initial round trip, these round trips can be performed in parallel, which reduces the overall resolution time [19]; parallel ARRF therefore needs a minimum of two round trips [16]. The authors of [19] also point out limitations: middleboxes that inspect resource record types might reject RRFRAGs, packet loss was not evaluated, and nothing prevents an on-path attacker from changing the RRSIZE fields in the initial response, so that the requestor allocates memory for the intermediate state and can only detect the attack when verifying the signature [19].

3) DNS over QUIC (DoQ): A third alternative is to move the DNS exchange entirely onto DoQ, which carries DNS messages over dedicated, encrypted QUIC connections instead of plain UDP or TCP [20]. Because QUIC implements its own stream-based delivery, loss recovery, and connection migration logic on top of UDP, large DNSSEC responses can be delivered reliably without being exposed to IP-layer fragmentation in the same way as plain UDP traffic. This reliability, however, is obtained at the cost of a QUIC connection and TLS handshake, which—analogous to the TCP case—adds round-trip overhead on a cold connection, while a warm, already-established DoQ connection can amortize this cost across multiple subsequent queries. This project evaluates DoQ as a time-permitting extension once the TCP-fallback and, where feasible, ARRF-based evaluations have been completed. RFC 9250 explicitly names as a design goal a transport that does not impose path MTU limitations on the size of DNS responses; DoQ ignores the EDNS(0) UDP payload size parameter, always assumes a maximum message size of 65,535 bytes, and uses the dedicated UDP port 853 [20]. Because the QUIC handshake combines the negotiation of cryptographic and transport parameters [29], a new DoQ connection needs one handshake round trip before the query is sent, i.e., two round trips in total, compared with three for DNS over TLS (DoT) with TLS 1.3 [26]. On resumed connections, clients may use 0-RTT data, but only for replayable transactions such as standard queries [20]. In the measurements reported in [26], DoQ outperformed DoT and DNS over HTTPS (DoH) in response time, but about 40% of DoQ handshakes required an additional round trip because resolvers enforced QUIC's anti-amplification limit [26], under which a server may send at most three times as many bytes as it has received from a not-yet-validated address [29].

4) Comparison of Transport Mechanisms: Fig. 3 contrasts the message flows of the mechanisms discussed above for a single query. On a new connection, DNS over UDP needs one round trip [26]; TCP fallback costs one wasted UDP round trip followed by at least two round trips over TCP [16]; parallel ARRF needs a minimum of two round trips [16]; and DoQ needs one handshake round trip followed by the query, i.e., two round trips [26], whereas clients and servers should reuse a single QUIC connection for multiple queries, so that the handshake is not repeated [20]. These counts ignore processing and serialization delays as well as additional round trips caused by packet loss or by exceeding the initial congestion window (Section II-D3).

Table II summarizes the main properties of each mechanism, including two further UDP-based proposals discussed in Section III: QNAME-Based Fragmentation (QBF) and TurboDNS.

III. State of the Art and Related Work

The transition of DNSSEC to PQC represents a fundamental protocol-redesign problem rather than a simple algorithm substitution. Standardized post-quantum signature schemes such as ML-DSA-44 (2,420-byte signatures) and SLH-DSA inherently exceed the operational 1,232-byte UDP payload threshold, triggering significant transport and performance degradation across the DNS ecosystem. Recent literature has explored several transport mechanisms, structural optimizations, and prototype implementations to evaluate and mitigate these constraints.

The authors of [21] describe their work as the first study of the applicability of quantum-safe signature algorithms to DNSSEC; they evaluated the third-round candidates of the NIST post-quantum standardization process against DNSSEC-specific requirements (Section II-C3). Only three algorithms—Falcon-512, RedGeMSS128, and Rainbow-Ia—partially met these requirements; the latter two have very small signatures but public keys of 375 kB and 158 kB, respectively [21]. The authors concluded that no algorithm fits all requirements perfectly and proposed protocol adaptations, namely increased TCP support and out-of-band key distribution [21]. The situation has changed since then: NIST did not select Rainbow for standardization after an attack recovered private keys for security category 1 parameters in a little over two days on a single laptop, and GeMSS was eliminated after an attack dramatically reduced its security [31]. Of the three algorithms identified in [21], only Falcon was selected for standardization, together with CRYSTALS-Dilithium and SPHINCS+ [31].

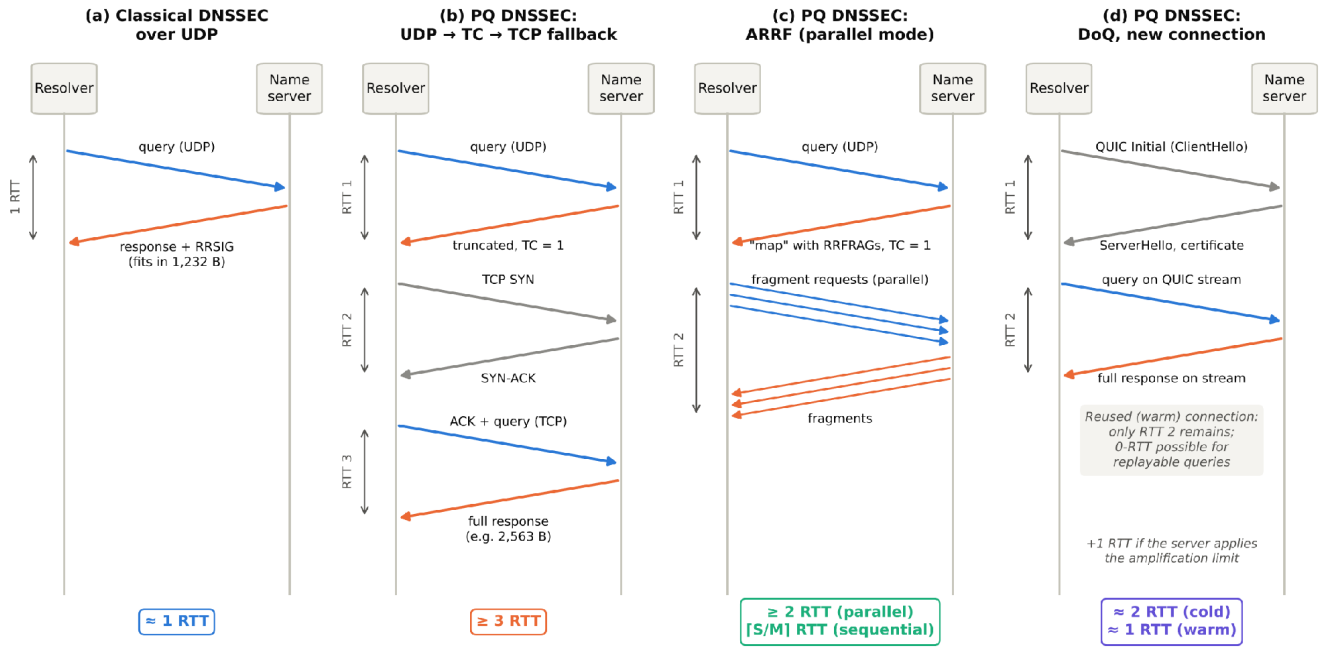


Fig. 3. Simplified message sequence diagrams for (a) classical DNSSEC over UDP, (b) post-quantum DNSSEC with truncation and TCP fallback, (c) ARRF in parallel mode, and (d) DoQ on a new connection.

TABLE II
TRANSPORT MECHANISMS FOR LARGE POST-QUANTUM DNSSEC RESPONSES

Mechanism	**Round trips (new connection)**	**Avoids IP fragmentation**	**Changes required**	**Main drawback**
UDP + EDNS(0) (classical) [1], [12], [23]	1	Only if the response fits the path MTU	None	Response is truncated (TC = 1) once it exceeds the payload limit
UDP → TC → TCP fallback [14], [16]	1 wasted (UDP) + ≥ 2 (TCP)	Yes (TCP stacks generally implement Path MTU Discovery)	None (TCP support already required)	Extra round trips and connection state
ARRF (parallel) [19]	≥ 2	Yes (fragments ≤ M)	Resolver and name server (RRFRAG)	Possible rejection by middleboxes; RRSIZE manipulation by on-path attackers
QBF (parallel 1-RTT) [30]	≈ 1	Yes	Resolver- and server-side daemon	Fragment count estimated in advance; no loss evaluation
TurboDNS [16]	Removes 2 RTTs from TCP fallback	Yes (TCP)	Resolver and name server (TD-Cookie)	Must be deployed on both resolvers and name servers
DoQ [20], [26], [29]	≈ 2 (cold), ≈ 1 (warm)	Yes (no path-MTU limit on DNS messages)	DoQ support on both sides (UDP port 853)	Handshake; amplification limit may add 1 RTT

Standard DNS over UDP relies on EDNS(0) buffer negotiation, but oversized post-quantum responses force authoritative servers to truncate messages and instruct resolvers to fall back to TCP [14]. Cloudflare's real-world deployment of ML-DSA-44 validation on its 1.1.1.1 resolver demonstrated that TCP fallback handles large DNSKEY responses reliably, albeit at the cost of additional RTT latency [11]. To optimize connection establishment overhead, the authors of [16] introduced TurboDNS, which accelerates TCP fallback by reusing stateless cookies (TD-Cookies) to bypass unnecessary initial handshake steps and reduce resolution latency [16]. In parallel, single-round-trip UDP fragmentation mechanisms such as QNAME-Based Fragmentation (QBF) [30] have been proposed to eliminate TCP connection setup entirely, achieving resolution times of about 45 ms at 10 ms one-way latency without packet loss [16]. In the TurboDNS evaluation, the mean resolution time of a Falcon-512-signed A query decreased from 89 ms with standard TCP fallback to 45 ms at 10 ms one-way latency, practically matching the 44 ms of ECDSA P-256 over UDP; QBF achieved comparable times [16]. Fig. 4 shows the corresponding mean resolution times reported in [16] for both network settings; Dilithium-2 there denotes CRYSTALS-Dilithium, on which ML-DSA is based [9]. QBF itself fragments and reassembles large post-

quantum DNSSEC messages over UDP using only standard DNS records: the requestor obtains the remaining fragments with specially formed query names, and in its parallel 1-RTT mode it requests all fragments at once [30]. Building on QBF-style fragmentation, the authors of [32] combine a classical and a post-quantum signature in each response, so that a response is accepted only if both verify, and report that this adds less than 9% to the mean resolution time compared with post-quantum signatures alone [32].

To avoid TCP connection setup overhead while remaining over UDP, the authors of [19] introduced ARRF, which fragments oversized DNSSEC records at the application layer rather than relying on IP-layer fragmentation [19]. However, security evaluations have highlighted significant vulnerabilities in application-layer fragmentation: the authors of [33] analyzed a memory-exhaustion vulnerability in ARRF and proposed a mitigation with minimal performance cost [33], while the authors of [34] addressed fragment mis-association and spoofing threats by introducing cryptographic commit-and-reveal chains [34]. Alternatively, transport protocols like DoQ [20] encapsulate DNS traffic within encrypted QUIC streams, eliminating IP fragmentation issues entirely through native stream multiplexing and congestion control. Quantitatively, the authors of [19] report that parallel ARRF was about 20% faster than TCP fallback; for Dilithium-2, for example, the resolution time decreased from 82.24 ms to 62.52 ms at 10 ms latency and 50 MB/s, and from 802.4 ms to 601.7 ms at 100 ms latency [19]. ARRF also transferred slightly less data than TCP fallback for Falcon-512 and Dilithium-2, whereas for SPHINCS+ TCP was more data-efficient because of the per-fragment RRFRAG overhead [19]; packet loss was not evaluated [19]. For DoQ, the authors of [26] measured a median handshake time of 235 ms compared with 156 ms for DNS over TCP and about 322 ms for DoT and DoH, from a single vantage point and without DNSSEC-specific workloads [26].

Recent empirical research has focused on prototype implementations and algorithm trade-offs. The authors of [17] extended CoreDNS with an open-source plugin supporting multiple post-quantum signature schemes—including ML-DSA, Falcon, SPHINCS+, MAYO, and SNOVA—providing benchmark evaluations of signing overhead and CPU utilization [17]. Their findings indicate that while Falcon-512 yields superior transport performance due to compact signatures, ML-DSA-44 offers lower computational complexity during key generation and signing [17]. SPHINCS+ presents the largest transport overhead, exceeding the initial congestion window (initcwnd) and requiring additional TCP round trips [16]. Similarly, the authors of [35] introduced a differentiated role-based algorithm assignment scheme for ZSKs and KSKs to optimize public key and signature size trade-offs [35], while Verisign and IETF proposals have explored Merkle Tree Ladder (MTL) modes to amortize signature sizes across multiple DNS responses [36]. The authors of [35] evaluated 18 configurations on a containerized testbed validated against the structure of AFNIC's `.fr` top-level domain and report that differentiated KSK/ZSK configurations add about 1.28 to 1.52 times the latency of classical ECDSA, while hybrid concatenation of classical and post-quantum signatures adds roughly 7% to 19% overhead [35]. The MTL mode was introduced in [37], whose authors show that, in a scenario with an initial series of 10,000 signatures, condensed signatures of 248–472 bytes can replace NIST PQC signatures of 666–49,856 bytes, depending on the security level [37]; an Internet-Draft by Verisign Labs specifies its use with SLH-DSA in DNSSEC, where resolvers request the larger full signatures only occasionally and preferably over TCP, TLS, or QUIC [36].

Table III summarizes the experimental work discussed in this section. Two observations emerge. First, most studies compare a newly proposed mechanism against plain TCP fallback, but under different testbeds, software versions, and network settings, which makes their absolute numbers difficult to compare. Second, the evaluations of ARRF and QBF explicitly do not consider packet loss [19], [30], and DoQ has so far been measured as an encrypted DNS transport rather than as a carrier for post-quantum DNSSEC [26]. A controlled comparison of TCP fallback, ARRF, and DoQ for ML-DSA-44 against an ECDSA baseline under varying latency and packet loss, as pursued in this project, therefore complements the existing literature.

IV. Conclusion of the Theoretical Part

This theoretical part has shown that the migration of DNSSEC to post-quantum signatures is, to a large extent, a transport problem. ML-DSA-44, the parameter set of the NIST standard FIPS 204 that is most suitable for use in the DNS, uses 1,312-byte public keys and 2,420-byte signatures [9], [10], so signed answers easily exceed the 1,232-byte EDNS(0) recommendation and the typical Internet MTU, as Cloudflare's 2,563-byte example shows [11]. Such responses cannot be delivered reliably through IP fragmentation [15], [23], [25]; they must instead be truncated and retried over TCP, fragmented at the application layer, or carried over a stream-based transport such as DoQ. At the same time, larger responses increase the potential for DNS amplification attacks [21], [27].

The literature reviewed in Section III shows that these options differ mainly in the number of round trips they require: for a single query on a new connection, TCP fallback costs one wasted UDP round trip plus at least two TCP round trips, parallel ARRF at least two round trips, and DoQ two round trips, with the handshake not repeated on a reused connection (Section II-E4) [16], [20], [26]. Published measurements show that standard TCP fallback roughly doubles the resolution time compared with classical DNSSEC over UDP, while under lossless conditions TurboDNS and QBF almost close this gap and parallel ARRF reduces it [16], [19], [30]. How these mechanisms behave under packet loss, and how ML-DSA-44 specifically compares with an ECDSA baseline in a reproducible setting, remains largely open.

These findings translate directly into the design of the practical part (Project – Part 2). Part 2 will address the research questions formulated in Section I: it will describe

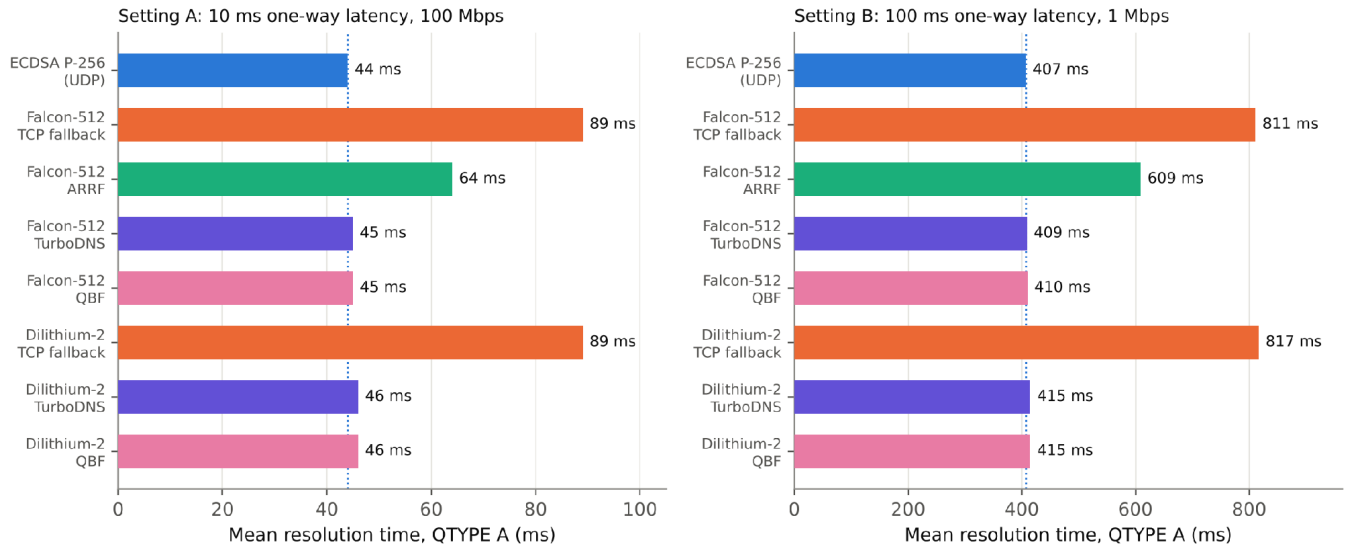


Fig. 4. Mean resolution times for QTYPE A queries for different signature algorithms and transport mechanisms under two network settings, as reported in [16]. The dotted line marks the ECDSA P-256 baseline.

TABLE III
OVERVIEW OF RELATED EXPERIMENTAL WORK ON POST-QUANTUM DNSSEC TRANSPORT

Mechanism	**Algorithms**	**Setting**	**Key result**
Requirements analysis [21]	NIST round-3 candidates	Analytical evaluation and benchmarks	No algorithm fits all requirements; three partially suitable (Rainbow, GeMSS later broken [31])
ARRF vs. TCP fallback [19]	Falcon-512, Dilithium-2, SPHINCS+	BIND 9 + daemon, Docker; 10/100 ms; packet loss not evaluated	Parallel ARRF $\approx 20\%$ faster than TCP fallback
QBF [30]	Falcon-512, Dilithium-2, SPHINCS+	Docker; 50 Mbps, 10 ms; no loss experiments	Parallel 1-RTT QBF $\approx 50\%$ faster than standard DNS
TurboDNS [16]	Falcon, Dilithium, SPHINCS+	10 ms/100 Mbps and 100 ms/1 Mbps	Falcon-512: 89 ms $\rightarrow$ 45 ms (ECDSA: 44 ms)
On-the-fly signing in CoreDNS, TCP [17]	13 PQC variants incl. ML-DSA-44	CoreDNS plugin	All PQC responses exceed UDP limits; $\approx$ 3–4 MB extra memory
Differentiated KSK/ZSK, hybrid signatures [35]	18 configurations	Testbed validated against the `.fr` TLD	1.28–1.52$\times$ ECDSA latency; hybrid +7–19%
Double signatures + fragmentation [32]	ECDSA/RSA + Falcon/Dilithium/ SPHINCS+	Docker testbed on Amazon EC2	$< 9\%$ overhead vs. PQC-only
DoQ, no DNSSEC [26]	–	264 resolvers, single vantage point	DoQ faster than DoT/DoH; expected 1-RTT handshake
TCP fallback in production (Cloudflare) [11]	ML-DSA-44	Public resolver 1.1.1.1	Validation deployed; 2,563-byte example response via TCP
This project: TCP fallback (+ ARRF, DoQ)	ECDSA P-256 vs. ML-DSA-44	Controlled latency and packet loss	See Part 2

the experimental testbed, the simulated network conditions, and the measurement methodology in detail and compare the measured results with the findings summarized above.

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